

ZILAIR, Russian Federation

WMO: 350260

Lat: 52.239N

Lon: 57.452E

Elev: 521

StdP: 95.22

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.9	(c) -25.5	(d) -31.8	(e) 0.2	(f) -28.5	(g) -28.7	(h) 0.3	(i) -25.1	(j) 7.2	(k) -9.3	(l) 6.2	(m) -9.9	(n) 1.0	(o) 30	(p) 0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.2	17.3	28.4	17.0	26.7	16.4	18.9	26.8	18.2	26.0	17.4	24.7	2.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	12.3	21.6	15.3	11.6	20.8	14.5	10.9	20.4	55.9	26.7	53.3	26.0	50.7	24.8	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.0	(b) 5.9	(c) 5.2	(e) -32.8	(f) 32.9	(g) 4.0	(h) 1.7	(i) -35.6	(j) 34.1	(k) -38.0	(l) 35.1	(m) -40.2	(n) 36.1	(o) -43.0	(p) 37.3		
(a) 7.0	(b) 5.9	(c) 5.2	(e) -32.8	(f) 20.3	(g) 3.9	(h) 1.4	(i) -35.6	(j) 21.3	(k) -37.9	(l) 22.1	(m) -40.1	(n) 22.9	(o) -43.0	(p) 23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-13.4	-12.0	-5.6	4.1	11.9	16.2	18.2	17.0	10.6	3.3	-5.1	-11.4	(6)
(7)		DBStd	12.55	5.59	5.90	4.91	5.66	4.48	4.52	3.81	4.37	4.49	4.85	5.92	6.29	(7)
(8)		HDD10.0	3429	726	615	484	190	33	6	1	3	45	211	452	664	(8)
(9)		HDD18.3	5750	984	848	742	427	203	91	51	79	235	465	702	923	(9)
(10)		CDD10.0	837	0	0	0	13	91	193	253	219	63	5	0	0	(10)
(11)		CDD18.3	115	0	0	0	0	3	28	45	36	3	0	0	0	(11)
(12)		CDH23.3	1409	0	0	0	2	74	378	488	427	40	0	0	0	(12)
(13)		CDH26.7	376	0	0	0	0	6	109	125	129	7	0	0	0	(13)
(14)	Wind	WSAvg	2.0	1.8	2.2	2.4	2.4	2.2	2.0	1.9	1.8	1.7	2.0	2.0	1.9	(14)
(15)	Precipitation	PrecAvg	518	39	36	46	43	41	47	49	40	35	48	49	46	(15)
(16)		PrecMax	977	96	98	152	108	95	116	104	170	99	106	106	102	(16)
(17)		PrecMin	332	0	3	0	4	4	0	8	4	2	0	2	2	(17)
(18)		PrecStd	132	24	23	33	29	26	27	30	34	27	27	28	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.6	1.0	9.1	22.7	27.3	31.8	32.4	32.4	27.3	19.2	7.9	0.6	(19)
(20)			MCWB	-1.0	-1.0	3.5	11.5	14.8	17.4	17.9	17.8	14.1	10.5	6.3	-0.4	(20)
(21)		2%	DB	-2.4	-0.9	5.4	19.8	25.3	29.6	29.7	30.1	23.8	16.3	5.4	-0.7	(21)
(22)			MCWB	-3.0	-2.0	1.6	10.4	14.0	17.2	17.8	17.1	13.3	8.8	4.0	-1.4	(22)
(23)		5%	DB	-3.5	-2.2	2.8	16.3	23.6	27.6	27.9	28.0	21.3	13.6	3.2	-1.7	(23)
(24)			MCWB	-4.0	-3.1	0.2	8.7	13.6	16.7	17.4	16.7	12.6	7.6	2.2	-2.3	(24)
(25)		10%	DB	-5.1	-3.6	1.1	13.1	21.3	25.2	26.1	25.7	18.7	11.1	1.7	-3.0	(25)
(26)			MCWB	-5.6	-4.5	-0.7	7.2	12.4	15.8	16.8	16.3	11.6	6.8	0.7	-3.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	-0.3	4.6	12.4	16.7	19.2	20.3	19.5	15.9	11.6	6.8	-0.1	(27)
(28)			MCDWB	-0.8	0.3	8.4	20.8	23.8	27.7	26.9	27.5	23.7	16.7	7.3	0.2	(28)
(29)		2%	WB	-2.8	-1.8	2.0	10.7	15.3	18.2	19.3	18.3	14.4	9.9	4.2	-1.3	(29)
(30)			MCDWB	-2.5	-1.1	4.7	18.0	22.9	27.0	26.6	26.6	21.1	14.6	5.1	-0.9	(30)
(31)		5%	WB	-3.9	-3.0	0.7	9.3	14.3	17.4	18.6	17.6	13.3	8.4	2.4	-2.2	(31)
(32)			MCDWB	-3.5	-2.3	2.4	15.5	21.9	25.6	25.8	25.7	19.2	12.5	3.2	-1.7	(32)
(33)		10%	WB	-5.6	-4.5	-0.4	7.7	13.2	16.5	17.7	16.8	12.2	7.1	0.9	-3.6	(33)
(34)			MCDWB	-5.2	-3.6	1.0	12.6	19.9	23.8	24.3	24.3	17.7	10.5	1.5	-3.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	9.1	9.0	9.9	12.4	12.3	11.7	12.1	11.2	8.4	5.7	7.1	(35)
(36)			MCDBR	5.5	9.4	10.7	14.9	16.0	15.6	14.4	15.3	15.9	14.4	6.9	6.0	(36)
(37)			MCWBR	5.4	7.9	7.5	7.6	6.8	5.9	5.4	5.2	7.3	8.4	5.6	5.5	(37)
(38)		5% WB	MCDWB	5.4	7.2	8.4	13.4	14.2	13.4	12.0	13.5	13.3	12.5	5.9	5.5	(38)
(39)			MCWBR	5.3	6.4	6.2	7.3	6.7	5.6	5.2	5.4	6.8	7.9	5.2	5.3	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.282	0.309	0.364	0.412	0.394	0.387	0.399	0.419	0.368	0.342	0.311	0.279	(41)	
(42)		taud	2.089	2.048	1.956	2.054	2.255	2.338	2.310	2.231	2.341	2.314	2.185	2.076	(42)	
(43)		Ebn at Noon	685	779	802	816	856	866	847	804	808	748	647	610	(43)	
		Edh at Noon	75	106	144	149	128	119	120	124	98	82	69	64		
(44)	All-Sky Solar Radiation	RadAvg	1.11	2.21	3.56	4.51	5.62	6.27	6.01	4.98	3.41	1.75	0.80	0.72	(44)	
(45)		RadStd	0.15	0.21	0.40	0.44	0.50	0.59	0.42	0.42	0.41	0.34	0.15	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+1.27	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page